

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is that the server had lost ability to respond because of the flooding SYN requests. The logs show that there has been an overwhelming amount of TCP/SYN requests from an unfamiliar IP address that have caused the server to stop responding. This event could be a SYN flood attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The handshake consists of these three steps:

1. The client sends an SYN request to a second party
2. The servers sends an SYN/ACK back to show it acknowledges and approves the request to exchange information
3. The message gets sent to the second party

When a malicious actor sends a large number of SYN packets all at once the server gets overwhelmed processing each request as it has to do them one by one until it just stops responding at all.

The logs indicate that an unfamiliar malicious IP address has flooded the server with SYN requests with the intention of stopping its response.